

Reproducibility Manifest

****Document ID:**** reproducibility-v1

****Date:**** 2026-08-31

1. Analysis Provenance

1.1 Analysis Identification

- ****Analysis ID:**** 228f930b076eb4c5495ac2b84066ba658b91a1c62a1302fc32a4f1bc96f8cdea
- ****Scientific Supplement ID:**** 3741598adc31cfdc61c050be5b97e2de93154823e6d859298456a73fb2d4125a
- ****Code Commit:**** See manifest.json

1.2 Data Sources

1.3 Configuration

- Analysis config: `config/analysis-plan-v1.yaml`
- Monitoring config: `config/monitoring-service-v1.yaml`
- Scientific config: `config/scientific-supplement-v1.yaml`

2. Execution Commands

2.1 Full Pipeline

```
```bash
docker compose run --rm --build pipeline
```
```

2.2 Analysis

```
```bash
docker compose run --rm analysis
```
```

2.3 Services

```
```bash
docker compose up --build evaluation-monitoring analytics-assistant
```
```

2.4 Tests

```
```bash
docker compose run --rm test
```
```

3. Verification

3.1 Hash Verification

All artifacts are content-addressed and hash-verified:

- 1,440 numeric cells registered
- 27 artifacts verified
- All SVG figures parse

3.2 Idempotent Rerun

Immediate rerun returns identical analysis ID.

4. Limitations

4.1 Sealed Scale

Sealed runtime and memory are not proven:

- Local data: ~1.9M rows, ~115 seconds
- Sealed: ~400M rows, expected <40 minutes

4.2 Docker Execution

Local Docker Linux engine was unavailable:

- Services cannot be run in current environment
- Code is verified but runtime is not

5. Dependencies

5.1 Python Packages

All dependencies are pinned in requirements.txt:

- duckdb
- pandas
- numpy
- scipy
- pyyaml
- pytest

5.2 System

- Python 3.12
- Docker
- Git

6. No Manual Calculation

****Every number in this analysis comes from a locked script.****

No notebook, spreadsheet, or manual calculation is in the production path.

This manifest documents the reproducibility of the locked analysis.